Object Oriented Analysis & Design

UE23CS352B

6th Semester, Academic Year 2023

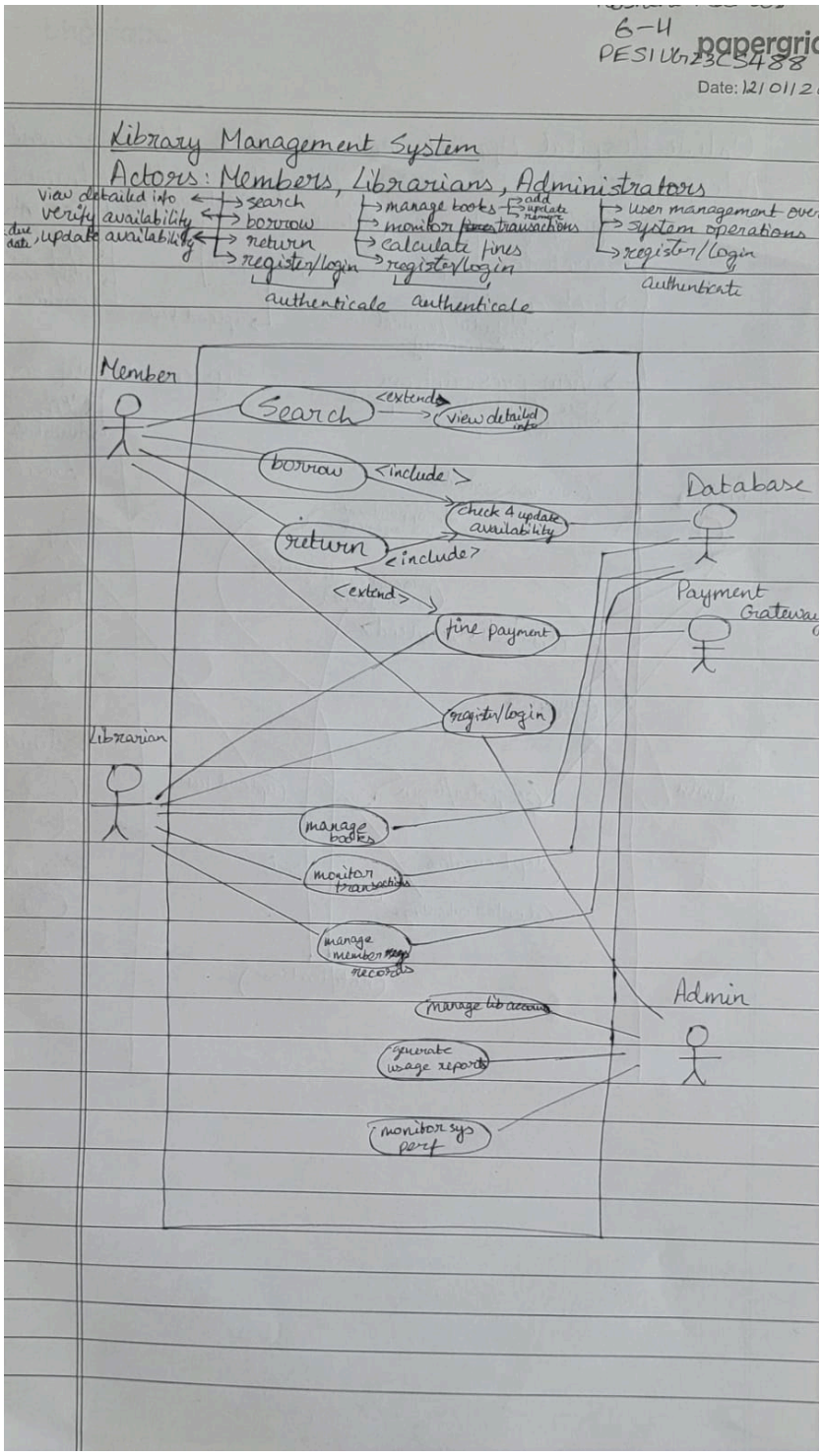
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Week 2: Case Study on Use Case Diagram

Case Study 1: Library Management System

Use Case Diagram:



Use Case-1

Name: Issue Book
Summary: The member selects a book to borrow. The system checks availability of the book and if the book is available, the member can borrow it and the status of the book is updated in the system.
Actors: Member, Database
Preconditions: The member must be logged in and registered. The book must be there in the database of books. The member shouldn't have any fines and the book must be available.
Description:

1. Member searches for the book
2. The member selects the book to be borrowed.
3. The system verifies the availability of the selected book.
4. The system changes the availability of the book to borrowed and calculates the due date.
5. The system generates a confirmation of the book being issued.

Exception:

1. Power Failure: If power fails during transaction recording stage, the data is not saved and the status of the book remains at available.
2. System Failure: If the system is not able to access the database during the transaction, the user is notified to try again later and no changes are made.

Alternative Flow:

1. Book Unavailable: If the book has been borrowed and not available, then the system denies the request and asks the member to place a reservation instead.

Post Conditions: The book status is updated to borrowed and the book is added to the member's current borrowed list.

Use Case-2

Name: Return Book
Summary: The member selects a book to return. The system checks if there are any overdue fees pending. If yes, the member is prompted to pay the fees. Once the fees are paid, the member can return the book and the status of the book is updated in the system.
Actors: Member, Database, Librarian
Preconditions: The member and librarian must be logged in and registered. The book must be there in the database of books and be marked as borrowed by the member.
Description:

1. Librarian enters the book id of the book to be returned
2. The system retrieves the records and checks current date with the due date.
3. The system processes the return.
4. The system changes the availability of the book to available.

Exception:

1. Power Failure: If power fails during transaction recording stage, the data is not saved and the status of the book remains at borrowed.

2. System Failure: If the system is not able to access the database during the transaction, the user is notified to try again later and no changes are made.
3. Invalid Book ID: If barcode is unreadable or is not exist, then the librarian must manually solve it.

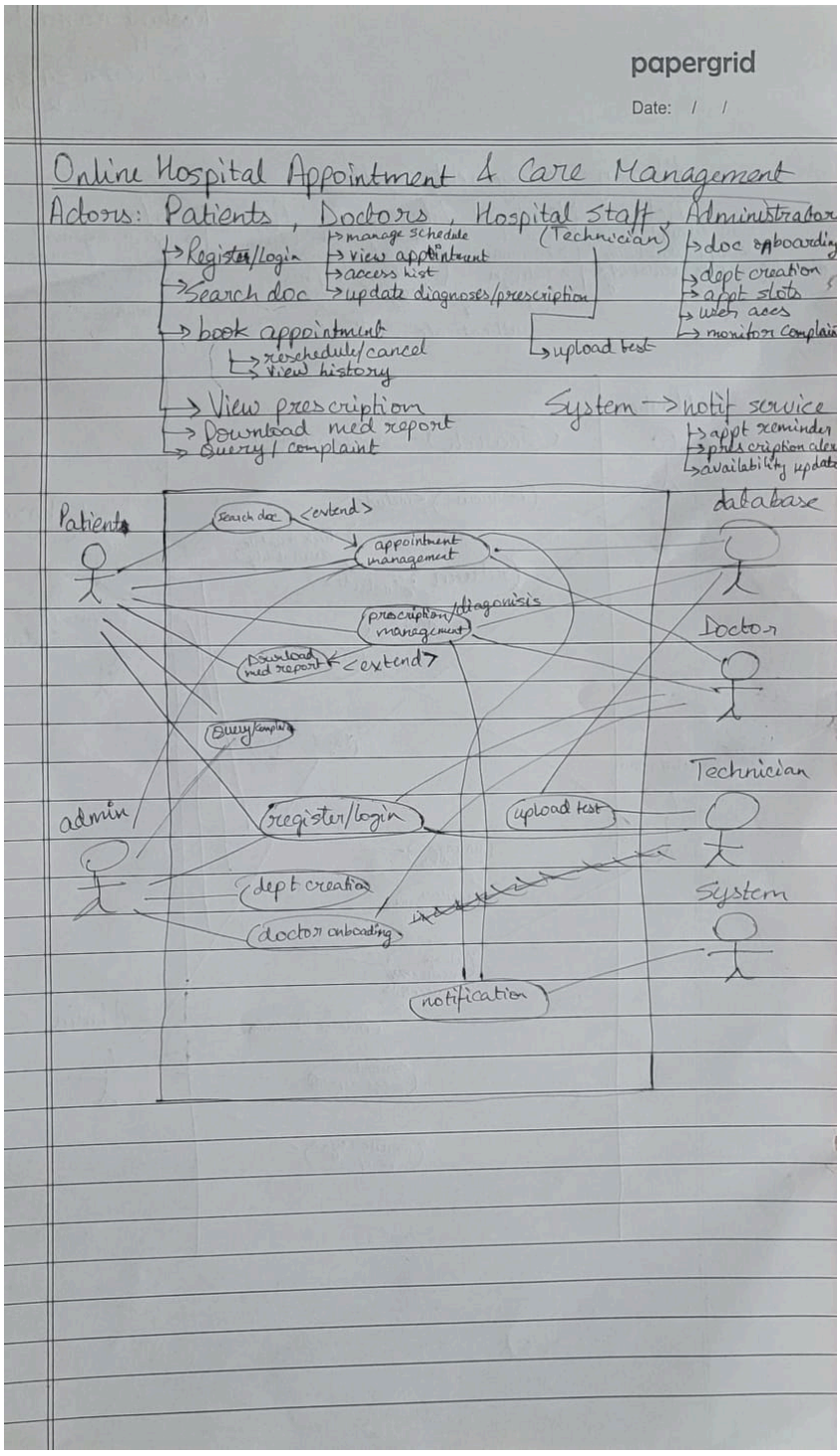
Alternative Flow:

1. Overdue Return: If return date is past due date, the librarian calculates the fine. The member is notified of the fine amount.
2. Damaged Book: If the Librarian flags the book as damaged, a replacement fee is calculated and charged on the member.

Post Conditions: The book status is updated to available and the book is removed from the member's current borrowed list.

Case Study 2: Online Hospital Appointment & Care Management

Use Case Diagram



Use Case-1

Name: Book Appointment

Summary: The patient searches for the doctor and schedules a consultation at a specific time.

Actors: Patient, System

Preconditions: The patient must be logged in and registered. The doctor must have available slots in the system.

Description:

1. Patient selects “Book Appointment”
2. Patient searches for their doctor by specialization, department, or availability.
3. The system displays a list of doctors with their available timeslot.
4. Patient selects doctor and timeslot.
5. The system verifies availability of the slot.
6. The system sends a confirmation and marks the doctor’s schedule as booked for the timeslot.

Exception:

1. Power Failure: If power fails during booking stage, the data is not saved and the status of the doctor for the timeslot remains available.
2. System Failure: If the system is not able to access the database during the transaction, the patient is notified to try again later and no changes are made.

Alternative Flow:

1. Cancellation: If the patient cancels the booking before confirmation, the doctor’s schedule is not booked and the system returns to the dashboard.

Post Conditions: The appointment is recorded in both the patient and the doctor’s schedule, timeslot is marked as booked and a notification is sent to the patient before the appointment.

Use Case-2

Name: Upload Test Results

Summary: A lab technician uploads the test report to the patient's profile.

Actors: Lab Technician, Database

Preconditions: The lab technician must be logged in and authenticated. The patient or the patient’s should have requested for a lab test and the results must be ready to be uploaded.

Description:

1. Lab technician searches for the patient using the patient ID.
2. The system checks if the patient exists.
3. The technician selects the test request.
4. Technician uploads the test report file.
5. The system links the report to the patient.
6. The system triggers a notification to the patient.

Exception:

1. Power Failure: If power fails during uploading stage, the data is not saved and the uploaded document is lost.
2. File Format Error: If the file is corrupted or of incorrect file format, the system rejects with an error message.

Alternative Flow:

1. Patient Not Found: If the patient ID doesn't match any existing patients, the system prompts the technician to reenter the ID or contact the admin.

Post Conditions: The medical report is ready for download by the patient and the patient's doctor can view it. The status of the test request is marked as complete.